

Lecture 10

The coordinate Bethe ansatz: the Heisenberg chain

Objectives. Course map: Part II begins; G2 and G3, step 1. *In 1931 Bethe wrote down eigenstates of the Heisenberg chain by a guess. Start from all spins up; flip one and it moves as a plane wave; flip two and they collide, and the collision does nothing but multiply the wave by a phase. Carry one flipped spin round the ring and the phases must add up: that condition, the Bethe equations, quantises the momenta. The lecture derives it by hand and writes it in the form the thermodynamic limit of [Lecture 16](#) uses. [Lecture 14](#) and [Lecture 15](#) recover the same Bethe equations and eigenvalues by an algebraic construction.*

10.1 The question

Part I found a classical many-body system's conserved quantities as the traces of one matrix, commuting because of one local identity. Part II builds the quantum counterpart: [Lecture 14](#) constructs commuting operators as the traces of one matrix again, and [Lecture 15](#) obtains the chain's Hamiltonian from their logarithmic derivative and diagonalises them. The classical count does not carry over, since for a finite quantum chain counting commuting operators decides nothing ([Lecture 0](#)'s caveat). It starts, as the subject did, with an exact solution found by a guess. So the question of this lecture is

how do you find the exact eigenstates of an interacting spin chain?

The answer is Bethe's ansatz: plane waves of flipped spins, whose only interaction is a two-body phase ([Proposition 10.1](#)), quantised by the Bethe equations ([Theorem 10.2](#)). The rule for passing from Part I to Part II comes first.

The rule of translation

Quantum mechanics keeps the shape of the classical equation of motion and changes its ingredients. States are vectors in a Hilbert space, observables are Hermitian operators, and in the Heisenberg picture an observable evolves by

$$\frac{dA}{dt} = \frac{1}{i\hbar} [A, H] . \quad (10.1)$$

Set against $\dot{f} = \{f, H\}$, this is the substitution $\{\cdot, \cdot\} \rightarrow [\cdot, \cdot] / i\hbar$ — checked on the one bracket that fixes it, $\{q, p\} = 1$ against $[q, p] = i\hbar$. An operator A is conserved exactly when $[A, H] = 0$, and two conserved operators are “in involution” when they commute. From here on $\hbar = 1$, except in [Lecture 14](#), where the classical limit $\hbar \rightarrow 0$ is taken and $\hbar$ is written.

10.2 The model

The spin- $\frac{1}{2}$ Heisenberg chain on a ring of N sites is

$$H = J \sum_{n=1}^N \left(\mathbf{S}_n \cdot \mathbf{S}_{n+1} - \frac{1}{4} \right) = \frac{J}{2} \sum_{n=1}^N (P_{n,n+1} - 1), \quad \mathbf{S}_{N+1} = \mathbf{S}_1, \quad (10.2)$$

with $P_{n,n+1}$ the permutation of the spins on sites n and $n+1$ (Definition 9.4). On the basis of up and down spins the bond term is simple: $\frac{J}{2}(P-1)$ gives zero on a parallel pair, and on an antiparallel pair it returns $\frac{J}{2}$ times (the swapped pair minus the pair itself).

Symmetries. The total spin $\mathbf{S} = \sum_n \mathbf{S}_n$ commutes with every $P_{n,n+1}$, since exchanging two spins does not change their sum; so H commutes with S^x, S^y, S^z — the $SU(2)$ symmetry — and with the shift U that moves every spin one site on. $S^z = \frac{N}{2} - M$, where M is the number of down spins, so H does not mix different M .

The reference state. In $|\uparrow\uparrow\rangle = |\uparrow\uparrow \cdots \uparrow\rangle$ every bond is parallel, so $H|\uparrow\uparrow\rangle = 0$. Since P has eigenvalues ± 1 , each bond term has eigenvalues 0 and $-J$. **The sign of J decides what $|\uparrow\uparrow\rangle$ is.** For $J < 0$ every bond term is ≥ 0 , so $H \geq 0$ and $|\uparrow\uparrow\rangle$ is a ground state: the *ferromagnet*, and the intuition of this lecture, where flipped spins are excitations that cost energy. For $J > 0$, the *antiferromagnet* and the course's frozen convention, $|\uparrow\uparrow\rangle$ is the highest level; for even N the ground state lies at $M = N/2$ (Lecture 16); for odd N the lowest level is made of spin- $\frac{1}{2}$ multiplets — two of them for $N = 3, 5, 7, 9$ (the check script) — with highest weights at $M = (N-1)/2$. Every formula below holds for either sign.

What must be found. Because H commutes with $SU(2)$, its eigenspaces are sums of $SU(2)$ multiplets, and each multiplet of total spin s contains exactly one *highest-weight* vector, killed by $S^+ = \sum_n S_n^+$, with $S^z = s$; the rest of the multiplet is got from it by applying S^- . Decomposing $(\mathbb{C}^2)^{\otimes N}$ (a prerequisite), the multiplets of spin $\frac{N}{2} - M$ number

$$\binom{N}{M} - \binom{N}{M-1}, \quad 0 \leq M \leq \frac{N}{2}. \quad (10.3)$$

So the task is: *find the highest-weight eigenvectors with M down spins, for each $M \leq N/2$.* The method looks for them as waves of flipped spins.

10.3 One magnon

Let $|x\rangle$ be the state with one down spin, at site x . Of the N bonds only the two touching x are antiparallel, so

$$H|x\rangle = \frac{J}{2} (|x-1\rangle + |x+1\rangle - 2|x\rangle), \quad (10.4)$$

a particle hopping on a ring. The plane wave $|k\rangle = \sum_x e^{ikx} |x\rangle$ is an eigenvector with

$$E(k) = J(\cos k - 1), \quad e^{ikN} = 1, \quad k = \frac{2\pi I}{N}, \quad I = 0, \dots, N-1. \quad (10.5)$$

This is a *magnon*, or spin wave: one flipped spin spread over the ring with momentum k . For the ferromagnet, $E = |J|(1 - \cos k) \approx \frac{1}{2}|J|k^2$ at small k : a quasiparticle with a quadratic dispersion.

Which are highest weight? $S^+|k\rangle = (\sum_x e^{ikx})|\uparrow\uparrow\rangle$, which vanishes for $k \neq 0$ and not for $k = 0$. And $|k=0\rangle = \sum_x |x\rangle$ is $S^-|\uparrow\uparrow\rangle$: not a new state, but the ferromagnetic multiplet one step down. So the $N-1$ magnons with $k \neq 0$ are the highest-weight states with $M = 1$, as (10.3) says.

The rapidity

From now on the momentum is traded for a *rapidity* λ ,

$$e^{ik} = \frac{\lambda + i/2}{\lambda - i/2}, \quad \text{i.e.} \quad \lambda = \frac{1}{2} \cot \frac{k}{2}, \quad E = -\frac{2J}{4\lambda^2 + 1}. \quad (10.6)$$

It maps $0 < k < 2\pi$ onto the real line, $k = \pi$ to $\lambda = 0$, and $k \rightarrow 0$ to $\lambda \rightarrow \pm\infty$. The reason for the change appears in the next section: the two-magnon phase depends on k_1 and k_2 in a complicated way, but on λ_1 and λ_2 only through their difference. **Three names for one number** — k , e^{ik} and λ — are a standard place to get lost; the lecture fixes λ here and keeps it.

10.4 Two magnons

The equations

Let $|x_1, x_2\rangle$, $x_1 < x_2$, have down spins at x_1 and x_2 , and look for an eigenvector whose amplitude on $|x_1, x_2\rangle$ is $\psi(x_1, x_2)$. For the moment think of a long open line and ignore the ends; the ring enters at the end of this section. There are two cases:

$$E\psi(x_1, x_2) = \frac{J}{2} \left[\sum_{\pm} (\psi(x_1 \pm 1, x_2) + \psi(x_1, x_2 \pm 1)) - 4\psi(x_1, x_2) \right] \quad (x_2 > x_1 + 1), \quad (10.7)$$

$$E\psi(x, x+1) = \frac{J}{2} [\psi(x-1, x+1) + \psi(x, x+2) - 2\psi(x, x+1)] \quad (x_2 = x_1 + 1). \quad (10.8)$$

In the bulk each down spin hops independently, with four antiparallel bonds. When the two are neighbours the bond between them is parallel: neither can hop onto the other, and there are only two antiparallel bonds (Figure 10.1).

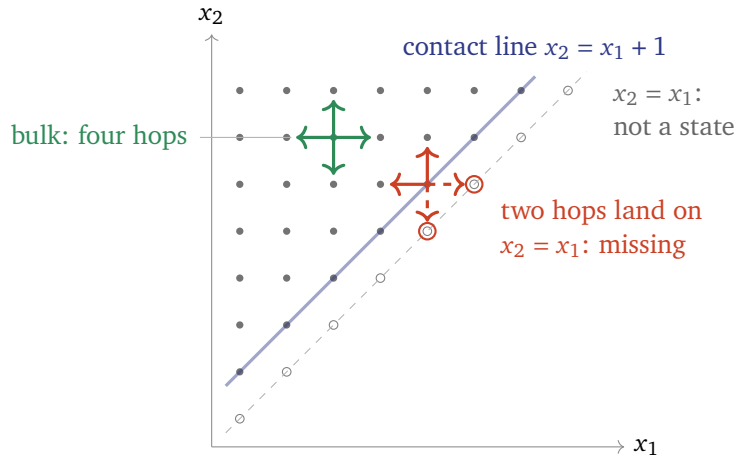


Figure 10.1: The two-magnon configurations $x_1 < x_2$ (dots). In the bulk (green) a configuration is joined to four neighbours. On the contact line $x_2 = x_1 + 1$ (red), two of the four hops would land on the diagonal $x_2 = x_1$, which is not a state; the true equation (10.8) lacks those two terms. The trick is to pretend they are there and demand that they cancel.

Bethe's ansatz and the contact condition

Two far-apart magnons should move as two plane waves, so try

$$\psi(x_1, x_2) = A_{12} e^{i(k_1 x_1 + k_2 x_2)} + A_{21} e^{i(k_2 x_1 + k_1 x_2)} ; \quad (10.9)$$

the two momenta, carried in either order. Each term solves the bulk equation (10.7) with the same energy

$$E = E(k_1) + E(k_2), \quad (10.10)$$

because in the bulk the two coordinates hop independently. The formula (10.9) defines ψ on the whole plane, the diagonal included, and satisfies the right-hand side of (10.7) at every point. Compare (10.8) with (10.7) evaluated at $x_2 = x_1 + 1$: the two differ by exactly the two missing hops, and the extra diagonal term,

$$(10.7) \text{ at } (x, x+1) - (10.8) = \frac{J}{2} [\psi(x+1, x+1) + \psi(x, x) - 2\psi(x, x+1)].$$

So the ansatz solves the true equations if and only if the *contact condition*

$$\psi(x, x) + \psi(x+1, x+1) - 2\psi(x, x+1) = 0 \quad \text{for all } x \quad (10.11)$$

holds. It concerns values of ψ at configurations that do not exist; that is the point of the trick.

Proposition 10.1 (The two-magnon S-matrix). *The ansatz (10.9) satisfies the contact condition if and only if*

$$\frac{A_{21}}{A_{12}} = S(\lambda_1 - \lambda_2), \quad S(\lambda) = \frac{\lambda - i}{\lambda + i}, \quad (10.12)$$

with λ_1, λ_2 the rapidities (10.6) of k_1, k_2 ; at $\lambda_1 - \lambda_2 = -i$, the pole of S , read it as $A_{12} = 0$.

Proof. Substitute: every term of (10.11) carries $e^{i(k_1 + k_2)x}$, and with $z_j = e^{ik_j}$ what remains is

$$A_{12} (1 + z_1 z_2 - 2z_2) + A_{21} (1 + z_1 z_2 - 2z_1) = 0.$$

Now put $z_j = (\lambda_j + i/2)/(\lambda_j - i/2)$ and clear the denominators $(\lambda_1 - i/2)(\lambda_2 - i/2)$: the first bracket becomes $\lambda_1 \lambda_2 - \frac{i}{2}(\lambda_1 + \lambda_2) - \frac{1}{4} + \lambda_1 \lambda_2 + \frac{i}{2}(\lambda_1 + \lambda_2) - \frac{1}{4} - 2(\lambda_1 \lambda_2 + \frac{i}{2}\lambda_1 - \frac{i}{2}\lambda_2 + \frac{1}{4}) = -i(\lambda_1 - \lambda_2 - i)$, and the second is the same with $1 \leftrightarrow 2$, namely $i(\lambda_1 - \lambda_2 + i)$. Divide. ■

Three properties of S carry the rest of the lecture. It depends on the rapidities only through $\lambda_1 - \lambda_2$, which is why rapidities were introduced. For real λ it is a phase, $|S(\lambda)| = 1$, and $S(\lambda)S(-\lambda) = 1$. And $S(0) = -1$: with equal rapidities $A_{21} = -A_{12}$ and $k_1 = k_2$, so the Bethe vector is zero — a Pauli principle that was not put in. (Whether a rescaled limit could still give a state belongs to the completeness question, a reading; roots are taken pairwise distinct from now on.) Physically, the two magnons meet, and the outgoing wave carries the *same two momenta*, exchanged, times a phase: no reflection into new momenta, and S is the entire effect of the collision. For two classical particles of equal mass that is forced by energy and momentum conservation (Proposition 0.1); for magnons it is not, at least at total momentum $K = \pi$, where $E(k_1) + E(k_2) = 2J(\cos \frac{K}{2} \cos \frac{k_1 - k_2}{2} - 1)$ is $-2J$ however K is split. Non-diffraction here is found by the ansatz, not supplied by kinematics.

The ring

On the ring, $|x_1, x_2\rangle$ with $1 \leq x_1 < x_2 \leq N$ is the same state as the down spins at x_2 and $x_1 + N$. Extend ψ by requiring

$$\psi(x_1, x_2) = \psi(x_2, x_1 + N); \quad (10.13)$$

then the equations of the ring are the equations of the line for $x_1 < x_2 < x_1 + N$, including the pair adjacent across the seam, $x_1 = 1, x_2 = N$. Substituting (10.9) and matching the coefficients of the two plane waves gives $A_{12} = A_{21}e^{ik_1N}$ and $A_{21} = A_{12}e^{ik_2N}$. With Proposition 10.1 these are the Bethe equations for two magnons; the reading at the end of the lecture states them as a theorem, and § 10.5 does M magnons at once.

10.5 M magnons

For M down spins at $x_1 < \dots < x_M$, Bethe's ansatz is a sum over all $M!$ ways of handing out the momenta:

$$\psi(x_1, \dots, x_M) = \sum_{\sigma \in S_M} A_\sigma \exp\left(i \sum_{j=1}^M k_{\sigma(j)} x_j\right), \quad A_\sigma = \prod_{\substack{a < b \\ \sigma \text{ puts } b \text{ before } a}} S(\lambda_a - \lambda_b), \quad (10.14)$$

where “ σ puts b before a ” means $\sigma^{-1}(b) < \sigma^{-1}(a)$; $A_{\text{id}} = 1$, and for $M = 2$ this is (10.12). The amplitude collects one factor of S for each pair of magnons whose momenta have been exchanged.

Theorem 10.2 (Bethe 1931 · sketched). Let $\lambda_1, \dots, \lambda_M$ be finite and pairwise distinct, none equal to $\pm i/2$, no two differing by $\pm i$, and let them satisfy the Bethe equations

$$\left(\frac{\lambda_j + i/2}{\lambda_j - i/2}\right)^N = \prod_{l \neq j} \frac{\lambda_j - \lambda_l + i}{\lambda_j - \lambda_l - i}, \quad j = 1, \dots, M. \quad (10.15)$$

Then (10.14), if it is not the zero vector, is an eigenvector of H with

$$E = - \sum_{j=1}^M \frac{2J}{4\lambda_j^2 + 1}, \quad S^z = \frac{N}{2} - M, \quad K = \sum_{j=1}^M k_j. \quad (10.16)$$

Sketch. Bulk. If no two down spins are neighbours, each plane wave describes M independent hoppers, and the energies add. *One contact.* If $x_{j+1} = x_j + 1$ and no other pair touches, the true equation differs from the extended bulk equation by $\frac{J}{2}$ times the contact expression (10.11) in the coordinates x_j, x_{j+1} , the others held fixed. Pair each σ with $\sigma' = \sigma \circ (j \ j+1)$, which exchanges the momenta in positions j and $j + 1$. Their two terms are a two-magnon wave in (x_j, x_{j+1}) times a common factor, so by Proposition 10.1 they cancel in the contact expression if $A_{\sigma'}/A_\sigma = S(\lambda_{\sigma(j)} - \lambda_{\sigma(j+1)})$ — and the product in (10.14) is built to do exactly that, gaining or losing the one factor for the pair exchanged (using $S(\lambda)S(-\lambda) = 1$). *Several contacts.* H is a sum of bonds; each missing hop belongs to one adjacent pair, so the true equation differs from the extended one by a sum of contact expressions, one per adjacent pair, each vanishing as before. A block of three neighbours needs nothing new. (The check script verifies the case $M = 3$ symbolically, three neighbours included.) *Periodicity.* The ring's condition is $\psi(x_1, \dots, x_M) = \psi(x_2, \dots, x_M, x_1 + N)$. In the right-hand side the term of σ is the plane wave in which x_1 carries k_a , $a = \sigma(M)$, followed by the others in the order

of σ ; call that ordering τ . So the condition reads $A_\tau = A_\sigma e^{ik_a N}$. Passing from σ to τ moves a from last to first, past every other magnon once: each pair $\{a, b\}$ changes its order, and by (10.14) $A_\tau/A_\sigma = \prod_{b \neq a} S(\lambda_b - \lambda_a)$. The condition is therefore $e^{ik_a N} = \prod_{b \neq a} S(\lambda_b - \lambda_a)$, which is (10.15). ■

Reading the Bethe equations. Write (10.15) as

$$e^{ik_j N} \prod_{l \neq j} S(\lambda_j - \lambda_l) = 1. \quad (10.17)$$

Carry magnon j once round the ring: it picks up $e^{ik_j N}$ from the distance travelled and one factor S from each other magnon it passes, and it must come back to the same wavefunction (Figure 10.2). The many-body problem has become M coupled equations for M numbers.

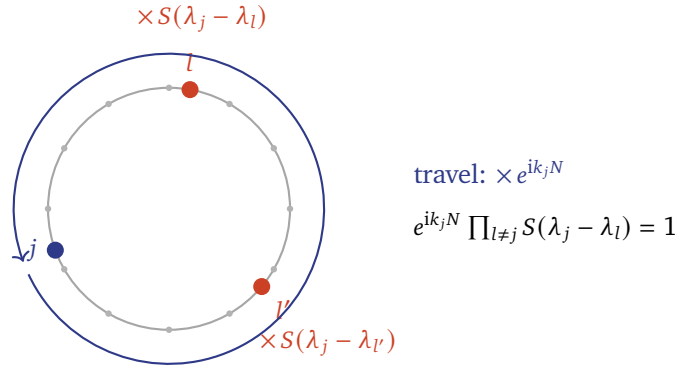


Figure 10.2: The Bethe equations as a round trip. Magnon j , carried once round the ring, gains the phase $e^{ik_j N}$ of the distance and one two-body phase from each other magnon it passes; the wavefunction must return to itself. Nothing depends on the order of the passes, because the S 's are numbers.

10.6 The logarithmic form

For real x let $\theta_n(x) = 2 \arctan(2x/n)$, so that $(x + in/2)/(x - in/2) = -e^{-i\theta_n(x)}$. Taking logarithms of (10.15) for real rapidities,

$$N\theta_1(\lambda_j) - \sum_{l \neq j} \theta_2(\lambda_j - \lambda_l) = 2\pi I_j, \quad I_j \in \mathbb{Z} + \frac{N-M+1}{2}, \quad (10.18)$$

the signs $(-1)^N$ and $(-1)^{M-1}$ having gone into I_j : the *Bethe quantum numbers* are integers if $N - M$ is odd and half-odd integers if $N - M$ is even. Summing (10.18) over j (the θ_2 's cancel in pairs, θ_2 being odd) and using $k = \pi - \theta_1(\lambda)$ gives the momentum

$$K = \pi M - \frac{2\pi}{N} \sum_j I_j \pmod{2\pi}. \quad (10.19)$$

As $\lambda_j \rightarrow \pm\infty$ the left side of (10.18) tends to $\pm(N - M + 1)\pi$, so $I_j = \pm \frac{N-M+1}{2}$ belongs to a root at infinity. The finite real roots of every solution met here have $|I_j| \leq \frac{N-M-1}{2}$, which leaves $N - M$ values (the limit explains the endpoint, not the bound). For $M = 2$ at $N = 6$ and 8, where every solution is known, each choice of two distinct values gives exactly one

real solution and there are no others — $\binom{N-M}{M}$ of them; at $N = 8, M = 3$, each of the $\binom{5}{3} = 10$ choices gives one (the check script; not proved in general). The I_j label states the way occupation numbers label free fermions. The tutorial at the end of this lecture draws them as the crossings of a *counting function*, the picture [Lecture 16](#) takes to $N \rightarrow \infty$.

Rewrite (10.17) with $S(\lambda) = -e^{i\theta_2(\lambda)}$:

$$Nk_j + \sum_{l \neq j} \theta_2(\lambda_j - \lambda_l) = 2\pi n_j + (M-1)\pi, \quad n_j \in \mathbb{Z}. \quad (10.20)$$

The $(M-1)\pi$ collects the $M-1$ minus signs of $S = -e^{i\theta_2}$. With one magnon the sum is empty and (10.20) is $kN = 2\pi n$: exactly the Bohr–Sommerfeld condition $\oint p \, dx = 2\pi\hbar n$ of a particle going once round a ring of circumference N (a reading of [Lecture 6](#)), and here it is not an approximation but the exact spectrum (10.5).

Highest weight

Theorem 10.3 (Bethe vectors are highest weight · cited). Under the hypotheses of [Theorem 10.2](#), if the vector (10.14) is not zero it is a highest-weight vector: $S^+\psi = 0$. Consequently $M \leq N/2$.

For $M = 1$ this is the computation of § 10.3: $S^+|k\rangle$ is $\sum_x e^{ikx}$, zero on the grid unless $k = 0$, i.e. unless $\lambda = \infty$. In general $S^+\psi$ is a sum of $(M-1)$ -magnon plane waves whose coefficients are, term by term, combinations of the Bethe equations, so vanish on solutions (Faddeev and Takhtajan 1981). The consequence follows because a highest-weight vector has $S^z = s \geq 0$.

10.7 What it buys

- *Exact eigenstates of an interacting chain*, labelled by M rapidities that solve M coupled equations ([Theorem 10.2](#)), with energy and momentum read off them (10.16).
- *Scattering that only exchanges momenta*. The whole effect of a collision is the phase $S(\lambda_1 - \lambda_2)$, and the M -magnon state is built from it alone: the second face of integrability that [Lecture 0](#) described, now for a quantum system.
- *Quantum numbers*. In logarithmic form each real root carries a quantum number I_j (10.18), as occupation numbers label free fermions; [Lecture 16](#) takes them to $N \rightarrow \infty$ as a density.

10.8 Caveats, in one place

- *Status*. [Proposition 10.1](#) is proved and [Theorem 10.2](#) is sketched, its contact book-keeping routine. That Bethe vectors are highest weight ([Theorem 10.3](#)) is cited.
- *Completeness*. Whether every highest-weight state is a Bethe state is not proved here. At $N = 6, M = 2$ all nine are found, six from real roots and three from complex or singular ones (the tutorial and the reading).
- *Real quantum numbers*. That each allowed set of distinct I_j gives a real solution is checked here only in small cases, not proved in general. At $N = 6$ and 8 with $M = 2$, where every solution is known, each set gives exactly one; at $N = 8$ with $M = 3$ each of the ten sets gives one, and uniqueness is not checked.

- The hypotheses of [Theorem 10.2](#) — finite, pairwise distinct rapidities, none equal to $\pm i/2$, no two differing by $\pm i$ — exclude the singular solutions, which the reading treats.
- An analogy, not a limit. At $M = 1$ the Bethe equation is Bohr–Sommerfeld; with M magnons the resemblance is structural. No $\hbar \rightarrow 0$ is taken, and magnons have no classical phase space.

Tutorial

10.9 A worked check, and how to reproduce it

	quantum numbers	roots λ_1, λ_2	E/J	K
real	$\pm \frac{1}{2}$	± 0.16246	-3.61803	0
real	$\pm \frac{3}{2}$	± 0.68819	-1.38197	0
real	$-\frac{3}{2}, \frac{1}{2} / -\frac{1}{2}, \frac{3}{2}$	$-0.63108, 0.19807 / \text{mirror}$	-2.5	$\pm\pi/3$
real	$-\frac{3}{2}, -\frac{1}{2} / \frac{1}{2}, \frac{3}{2}$	$-0.58200, -0.09417 / \text{mirror}$	-2.78078	$\pm 2\pi/3$
2-string	—	$\pm 0.55459 \pm 0.51247i$	-0.71922	$\pm 2\pi/3$
singular	—	$\pm i/2, \text{regularised}$	-1	π

Table 10.1: All nine highest-weight states of $N = 6$, $M = 2$. “Mirror” is $\lambda \rightarrow -\lambda$, which sends $K \rightarrow -K$. Every energy agrees with exact diagonalisation of the 15×15 block to 10^{-9} .

[Table 10.1](#) and [Figure 10.3](#) are the lecture checked end to end on the smallest interesting case. The Bethe solutions, and the 9 eigenvectors of H on the 15 states with two down spins that are killed by S^+ , agree in both directions, with every type of solution present (the tutorial says how to reproduce this). At $N = 8$ they agree again: 15 real solutions, 4 strings and the singular pair make the $\binom{8}{2} - 8 = 20$ highest-weight states (the check script). The same script finds the ground state of the whole $N = 8$ antiferromagnet as the Bethe state with $M = 4$ and $I_j = -\frac{3}{2}, \dots, \frac{3}{2}$, all real: the state [Lecture 16](#) takes to $N \rightarrow \infty$.

Reproducing the check. For $M = 2$ every solution can be found, not just the real ones. At each total momentum $K = 2\pi m/N$, write $e^{ik_2} = e^{iK} e^{-ik_1}$ and clear denominators in the first equation of (10.22): it becomes a polynomial of degree at most $N + 1$ in $z = e^{ik_1}$ (the degree drops at $K = \pi$),

$$-(w + 1)z^{N+1} + 2wz^N + 2z^2 - (w + 1)z = 0, \quad w = e^{iK},$$

with the roots $z = 0$ (the singular pair, whose partner $e^{ik_2} = w/z$ is infinite) and $z = 1$ (a root at infinity, a descendant). Discard both from the search for regular roots, and keep the others whose two rapidities are finite and distinct; for even $N \geq 4$ add the singular state by hand (for odd N , and $N = 2$, there is none). Separately, diagonalise H on the $\binom{N}{2}$ states with two down spins and keep the eigenvectors killed by S^+ , sorted by momentum. At $N = 6$ this is [Table 10.1](#).

Roots and holes. Define the counting function

$$\zeta(\lambda) = \theta_1(\lambda) - \frac{1}{N} \sum_l \theta_2(\lambda - \lambda_l). \quad (10.21)$$

The roots are the points where $N\zeta/2\pi$ passes one of its values I_j ; the allowed values it passes where there is no root are *holes* ([Figure 10.4](#)). In [Lecture 16](#), where $N \rightarrow \infty$, the densities

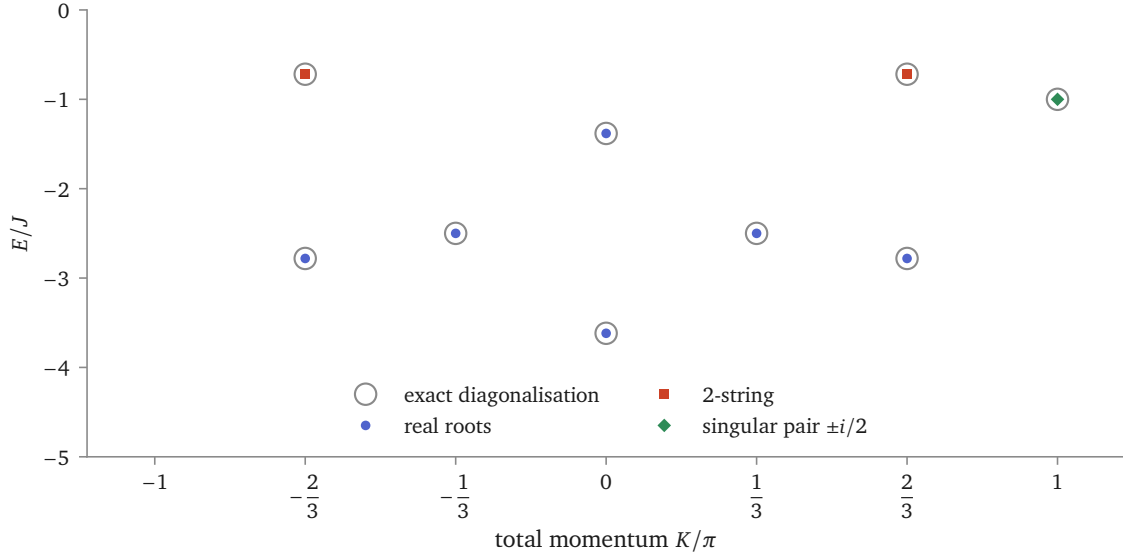


Figure 10.3: $N = 6$, $M = 2$: the nine highest-weight levels from exact diagonalisation (rings) and from the Bethe equations (dots), by type. The generator (`scripts/figures/xxx_bethe.py`) matches the two lists, energy and momentum, to 10^{-9} in both directions.

of roots and of holes add up to $\zeta'/2\pi$, and the ground state and its excitations are read off this picture.

The singular pair, regularised. Put $\lambda_1 = i/2 + \epsilon + c\epsilon^N$ and $\lambda_2 = -i/2 + \epsilon$, with $\epsilon \rightarrow 0$. Then $e^{ik_1} \approx i/\epsilon$ and $e^{ik_2} \approx i\epsilon$, while $\lambda_1 - \lambda_2 = i + c\epsilon^N$. In the first equation of (10.22) the left side is $\approx i^N \epsilon^{-N}$ and the right side $(2i + c\epsilon^N)/(c\epsilon^N) \approx 2i/(c\epsilon^N)$, so $c = 2i^{1-N}$. In the second the left side is $\approx i^N \epsilon^N$ and the right side $(-c\epsilon^N)/(-2i - c\epsilon^N) \approx c\epsilon^N/2i$, so $c = 2i^{N+1}$. The two agree if and only if $i^{2N} = 1$, that is, N even. With that c , in (10.9) the first wave is of order ϵ on nearest neighbours and smaller elsewhere, and the second, whose amplitude $S(\lambda_1 - \lambda_2)$ is of order ϵ^N , contributes at order ϵ only to the pair adjacent across the seam; divided by ϵ , the vector tends to a multiple of ψ_{sing} (the check script does this to 400 digits for $N = 4, \dots, 10$). The energy is finite: $-\sum_j 2J/(4\lambda_j^2 + 1) \rightarrow -J$, the two infinities cancelling.

Reading

10.10 Two magnons as a theorem; descendants; complex and singular roots

Theorem 10.4 (Two magnons). Let $\lambda_1 \neq \lambda_2$ be finite, neither equal to $\pm i/2$, with $\lambda_1 - \lambda_2 \neq \pm i$, and suppose

$$\left(\frac{\lambda_1 + i/2}{\lambda_1 - i/2}\right)^N = \frac{\lambda_1 - \lambda_2 + i}{\lambda_1 - \lambda_2 - i}, \quad \left(\frac{\lambda_2 + i/2}{\lambda_2 - i/2}\right)^N = \frac{\lambda_2 - \lambda_1 + i}{\lambda_2 - \lambda_1 - i}. \quad (10.22)$$

Then (10.9) with (10.12), if it is not the zero vector, is an eigenvector of H with $E = -\sum_j 2J/(4\lambda_j^2 + 1)$, $S^z = \frac{N}{2} - 2$, and momentum $K = k_1 + k_2$: $\psi(x_1 + 1, x_2 + 1) = e^{iK}\psi(x_1, x_2)$.

Proof. By Proposition 10.1 and the reduction above, the ansatz solves the equations of the

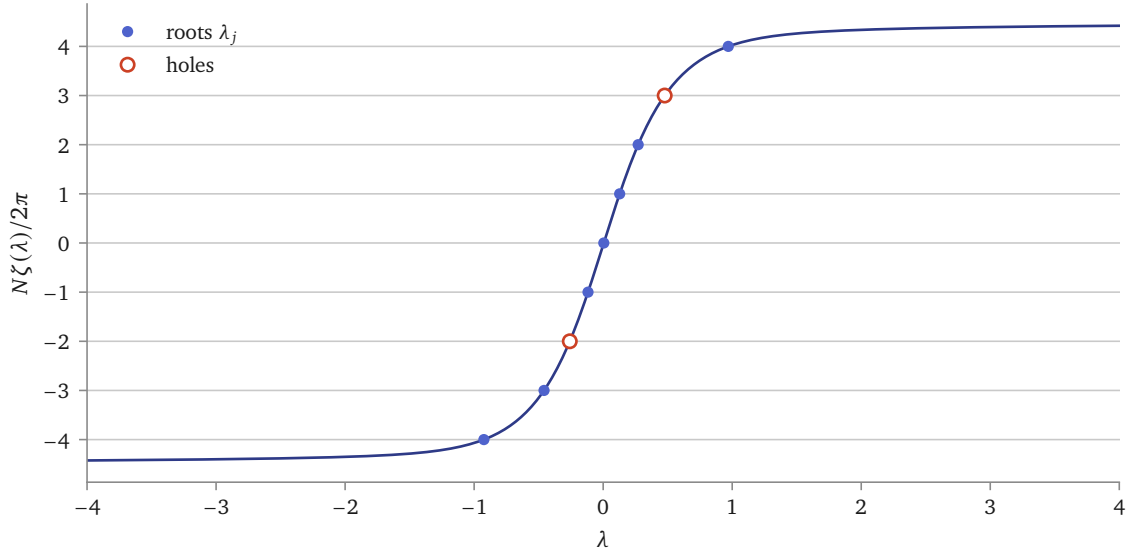


Figure 10.4: The counting function (10.21) for a state with $N = 16$, $M = 7$: the I_j fill seven of the nine allowed integers $|I| \leq 4$ and leave holes at -2 and 3 . Roots and holes sit where the staircase of allowed values meets the curve; as $\lambda \rightarrow \pm\infty$ it tends to $\pm \frac{N-M+1}{2} = \pm 4.5$, which is why $|I_j| \leq 4$. (scripts/figures/xxx_bethe.py, which asserts the equations to 10^{-10} and that the curve increases.)

line; by (10.13) it solves those of the ring provided $e^{ik_1 N} = A_{12}/A_{21} = 1/S(\lambda_1 - \lambda_2)$ and $e^{ik_2 N} = S(\lambda_1 - \lambda_2) = 1/S(\lambda_2 - \lambda_1)$, which are (10.22). The energy is (10.10) in rapidities, and the momentum is read off the plane waves. ■

Nothing here needed λ_j real. Multiplying the two equations gives $e^{iKN} = 1$: the total momentum is quantised as for one particle, and all the interaction sits in the relative motion.

The periodicity step is a complete argument; what the sketch leaves out is the bookkeeping of the contact steps, which is routine. One step deserves a flag. The amplitudes were defined by a closed formula, so they are consistent automatically; had we only known the two-body rule $A_{\sigma'}/A_{\sigma} = S$, we would have had to check that different sequences of exchanges leading to the same σ agree. For numbers that commute this is free. For particles carrying an internal label, S becomes a matrix, and the same requirement becomes the Yang–Baxter equation — Lecture 13’s subject.

The descendants need no new solutions, and one of them is a root at infinity. If $\lambda_2 \rightarrow \infty$ then $k_2 \rightarrow 0$, $S(\lambda_1 - \lambda_2) \rightarrow 1$, and (10.9) becomes $e^{ik_1 x_1} + e^{ik_1 x_2}$: exactly S^- applied to the one-magnon state $|k_1\rangle$, since S^- adds the second down spin at every other site. The equation for λ_1 becomes the one-magnon condition $e^{ik_1 N} = 1$. So the states of a multiplet below its top are S^- applied, repeatedly, to its highest-weight vector — one application being literally a Bethe state with one root at infinity — and (10.3) is the number of finite-root Bethe states the method must find. Whether it always finds them is the completeness question below.

For $N = 6$ and $M = 2$, (10.3) asks for $\binom{6}{2} - \binom{6}{1} = 9$ highest-weight states, and real roots supply $\binom{4}{2} = 6$. The other three are two strings and one singular solution.

Strings

Suppose $\text{Im } \lambda_1 > 0$. Then λ_1 is nearer to $i/2$ than to $-i/2$, so $|e^{ik_1}| > 1$ and the left side of (10.15) grows like $|e^{ik_1}|^N$. For large N the right side must follow, and it can only do so

through a nearly vanishing denominator: some λ_l with $\lambda_1 - \lambda_l \approx i$. The argument is heuristic (it assumes $\text{Im } \lambda_1$ stays away from 0 as N grows); for $M = 2$ at fixed total momentum it suggests

$$\lambda_{1,2} = x \pm \frac{i}{2} + O(\delta), \quad \delta \rightarrow 0 \text{ exponentially in } N, \quad (10.23)$$

a 2-string. In the wavefunction (10.9), $|e^{ik_1}| > 1 > |e^{ik_2}|$, so the wave $e^{i(k_2 x_1 + k_1 x_2)}$ grows with $x_2 - x_1$; but its amplitude $A_{21} = S(\lambda_1 - \lambda_2)A_{12}$ tends to zero as $\lambda_1 - \lambda_2 \rightarrow i$. What remains, $e^{iKx_1} e^{ik_2(x_2 - x_1)}$, decays in $x_2 - x_1$: a *bound state* of two magnons. For the exact string $\lambda = x \pm i/2$,

$$e^{iK} = \frac{x+i}{x-i}, \quad x = \cot \frac{K}{2}, \quad E = -\frac{J}{x^2 + 1} = \frac{J}{2}(\cos K - 1),$$

and for the ferromagnet this lies below the two-magnon continuum at the same K , whose bottom is $2|J|(1 - |\cos(K/2)|)$, by $|J|(1 - |\cos(K/2)|)^2$: two flipped spins side by side cost two broken bonds, two apart cost four. Figure 10.5 shows the roots of $N = 6$ and the decay of δ .

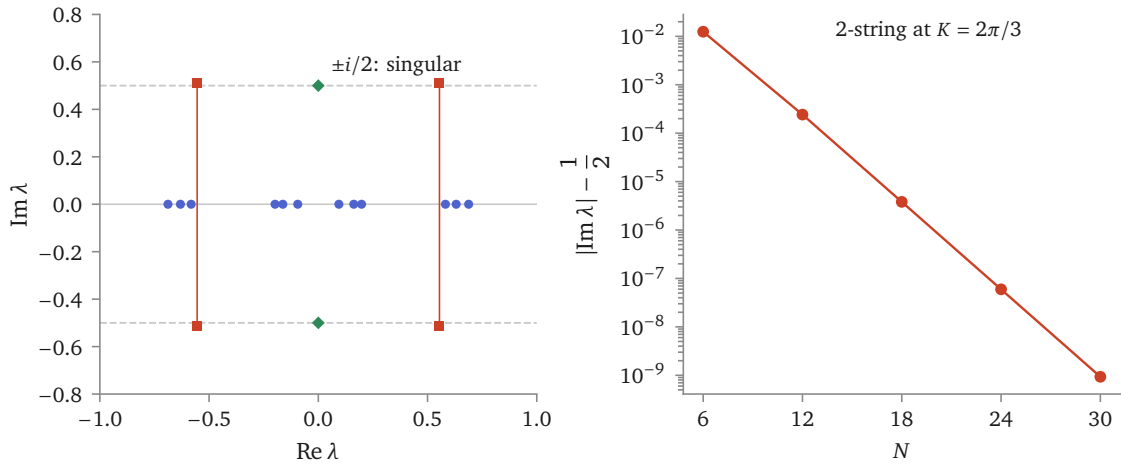


Figure 10.5: Left: the rapidities of all eight regular solutions with $N = 6$, $M = 2$ — six real pairs, two 2-strings (joined) near $x \pm i/2$ — and the singular pair $\pm i/2$. Right: the string at $K = 2\pi/3$ approaches $\text{Im } \lambda = \pm \frac{1}{2}$ exponentially in N , by a factor of about 50 for every six sites. (scripts/figures/xxx_bethe.py.)

For larger M the same argument makes complex roots cluster into strings of any length, $x + i(\frac{n+1}{2} - a)$, $a = 1, \dots, n$; that they *always* do, up to exponentially small corrections, is the *string hypothesis*, which is false in detail (Essler, Korepin and Schoutens 1992; there are “wide pairs” and collapsing strings) and still counts correctly (Lecture 17).

Highest weight, and roots at infinity

Singular solutions

The hypotheses of Theorem 10.2 exclude $\lambda = \pm i/2$. There e^{ik} is ∞ or 0, each term of the energy is infinite, and $\{i/2, -i/2\}$ satisfies the Bethe equations only in the degenerate sense $\infty = \infty$ and $0 = 0$. Nothing in the naive formulas makes sense, and yet something is there. Try

$$\psi_{\text{sing}} = \sum_{x=1}^N (-1)^x |x, x+1\rangle \quad (N \geq 4 \text{ even}),$$

a nearest-neighbour pair with alternating signs. By the rule of § 10.2, $H|x, x+1\rangle = \frac{J}{2}(|x-1, x+1\rangle + |x, x+2\rangle - 2|x, x+1\rangle)$; in the sum, $|y, y+2\rangle$ arrives from $x = y+1$ and from $x = y$ with opposite signs and cancels, leaving $H\psi_{\text{sing}} = -J\psi_{\text{sing}}$. It has $K = \pi$, and S^+ removes either down spin, giving $\sum_x (-1)^x (|x+1\rangle + |x\rangle) = 0$: a highest-weight eigenvector with $M = 2$. For odd N the signs clash across the seam and it is not an eigenvector. It is the missing ninth state at $N = 6$.

It is the singular pair, regularised: move the two roots off $\pm i/2$ by amounts that the Bethe equations themselves fix, and the naive vector, rescaled, tends to ψ_{sing} , while the two infinite energies cancel to $-J$. The two equations turn out to be compatible if and only if N is even (the tutorial at the end of this lecture does the computation). In general a solution containing the pair $\pm i/2$ gives an eigenvector only under an extra condition, and the solutions that do are called *physical* (Nepomechie and Wang 2013) · *cited*.

Completeness

Is every highest-weight state a Bethe state? Bethe counted: if every solution were made of strings, the solutions with M finite roots would number exactly (10.3). The count is right although the assumption is not. What is known · *cited*: numerically, for all $N \leq 14$, the solutions with finite, pairwise distinct roots and nonzero vectors, together with the physical singular ones, are exactly as many as (10.3) (Hao, Nepomechie and Sommesse 2013); a proof exists for a reformulation, a polynomial condition on Baxter's function Q of Lecture 15 that absorbs the singular cases (Mukhin, Tarasov and Varchenko 2009). For the equations as written, completeness is supported by numerics and counting, not proved. What the course uses is certain: every solution of the kind in Theorem 10.2 is an eigenstate.

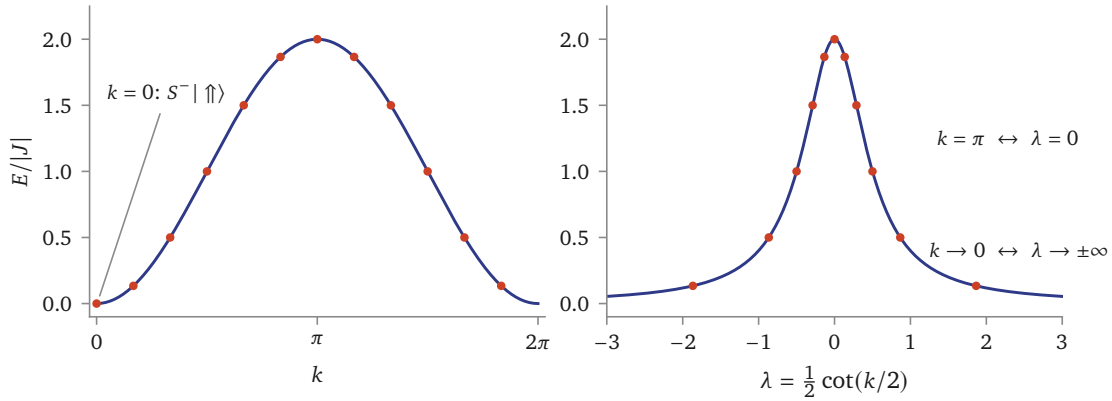


Figure 10.6: One magnon on a ring of $N = 12$ sites, for the ferromagnet $J < 0$. Left: $E = |J| (1 - \cos k)$ with the twelve allowed momenta; the dot at $k = 0$ is $S^- |\uparrow\rangle$, not a new state. Right: the same dots against the rapidity $\lambda = \frac{1}{2} \cot(k/2)$. The generator (`scripts/figures/magnon_dispersion.py`) asserts that the dots are the exact $M = 1$ spectrum.

10.11 Why Bohr–Sommerfeld works, and why the Bethe equations need no correction

Lecture 6 left two things to Part II: the correction $n \mapsto n + \frac{1}{2}$, and the reason (6.18) works at all. Both come from the WKB form of a wavefunction in one dimension,

$$\psi(q) \approx \frac{C}{\sqrt{|p(q)|}} e^{iS(q)/\hbar}, \quad S(q) = \int^q p \, dq', \quad p(q) = \sqrt{2(E - V(q))},$$

valid where the wavelength $2\pi\hbar/|p|$ changes little over one wavelength — that is, away from the turning points, where $p = 0$. Along a closed orbit ψ must come back to itself, which asks $\oint p \, dq/\hbar \in 2\pi\mathbb{Z}$: that is (6.18), and the reason it works is that $e^{iS/\hbar}$ is the leading behaviour of the true wavefunction as $\hbar \rightarrow 0$. Near a turning point the approximation fails, but there the potential is nearly linear and the exact solution is an Airy function; matching it to the WKB waves on either side (the *connection formulas*) shows that each reflection at a smooth turning point costs a quarter period, a phase $-\pi/2$ · *cited: Landau and Lifshitz, Quantum Mechanics, §47*. A libration has two turning points, so

$$\oint p \, dq = 2\pi\hbar(n + \tfrac{1}{2}).$$

For the oscillator, $I = E/\nu$ (§ 6.3) gives $E = \hbar\nu(n + \frac{1}{2})$, the exact spectrum with its zero-point energy. With n degrees of freedom, each basis cycle γ_i of the torus gets its own count μ_i of such phases, the *Maslov index*, and $\oint_{\gamma_i} p \cdot dq = 2\pi\hbar(n_i + \mu_i/4)$ (Einstein, Brillouin and Keller) · *cited*. A rotation, such as the pendulum over the top, has no turning point and no correction.

The Bethe equations have neither error. A magnon between collisions is not a WKB wave but an exact plane wave: the lattice has no potential, the momentum is conserved, and there is no turning point — motion round a ring is a rotation. The collisions contribute S , which is an exact solution of the two-body problem, not a semiclassical phase. So the round trip of Figure 10.2 is exact, and so is the analogy's $M = 1$ case.

Summary.

- *The model*: the spin- $\frac{1}{2}$ Heisenberg chain (10.2), with $SU(2)$ symmetry; the task is its highest-weight states, counted by (10.3).
- *One magnon*: a plane wave with $E = J(\cos k - 1)$; the rapidity $\lambda = \frac{1}{2} \cot(k/2)$ (10.6).
- *Two magnons*: the collision multiplies the wave by $S(\lambda_1 - \lambda_2) = (\lambda_1 - \lambda_2 - i)/(\lambda_1 - \lambda_2 + i)$ (Proposition 10.1), and nothing else.
- *M magnons*: the Bethe equations (10.15), a round trip of each magnon, and the energy (10.16); in logarithmic form, one quantum number per real root (10.18).
- *Next*: Lecture 11 solves a gas the same way, the Lieb–Liniger gas of the quantum Newton's cradle.

Tutorial. Exercise Sheet 10.

References

H. Bethe, Z. Phys. 71 (1931) 205; Faddeev and Takhtajan, Phys. Lett. A 85 (1981) 375; Essler, Korepin and Schoutens, J. Phys. A 25 (1992) 4115; Korepin, Bogoliubov and Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, ch. I.